Algebra 2 Review: Functions & Transformations

Answer Key

Station 1: Function Notation, Relations, Domain, and Range

1. 34
   $2(9) + 15 + 1$
2. $a^2 - 4$
   $(a+2)^2 - 4(a+2) = a^2 + 4a + 4 - 4a - 8$
3. A is a function: domain $\{1, 2, 3, 4\}$, range $\{3, 5, 7\}$. B is not: $x = 2$ gives both -1 and 3.
4. No: the vertical line $x = 2$ meets the graph at $(2, 2)$ and $(2, -2)$
   graph of $x = y^2 - 2$
5. $f(-1) = 4$, $f(2) = -2$; $x = -2,\ 1,\ 3$
   zeros are the x-intercepts
6. $\left(-\infty, \frac{5}{2}\right]$
   $5 - 2x \geq 0 \Rightarrow x \leq \frac{5}{2}$
7. $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$
   $x^2 - 9 \neq 0$*; the numerator may be* 0
8. Domain: all reals; range: $(-\infty, 3]$
   vertex $(2, 3)$*, opens down*
9. $C(120) = \$65$: 120 miles cost \$65. $m = 180$: \$80 pays for 180 miles.
10. $[-3, 1) \cup (1, \infty)$
    $x + 3 \geq 0$ *and* $x \neq 1$

Station 2: Parent Functions and Transformations of Graphs

1. Left 3, vertical stretch by 2, reflect in the x-axis, up 1; vertex $(-3, 1)$
2. Right 4, vertical compression by $\frac{1}{2}$, reflect in the x-axis, up 2; vertex $(4, 2)$; range $(-\infty, 2]$
3. $(-5, 1)$
   x *shifts LEFT 3:* $-2 - 3 = -5$*; y:* $5 - 4 = 1$
4. $(12, 2)$
   horizontal stretch by 2: $6 \cdot 2 = 12$*; reflect:* $-(-2) = 2$
5. $y = \sqrt{-x} - 1$; domain $(-\infty, 0]$; range $[-1, \infty)$
6. $y = -\frac{1}{2}|x - 2| + 3$
   vertex $(2, 3)$*; slope* $-\frac{1}{2}$ *on the right*
7. $y = 2\sqrt{x+1} - 3$
   starts at $(-1, -3)$*; passes through* $(3, 1)$
8. No: vertices $(0, 3)$ and $(0, 6)$
   stretch by 2 first and then shift up 3, versus shift up 3 first and then stretch
9. Horizontal stretch by 2, down 2; $g(4) = 6$
   $\left(\frac{1}{2} \cdot 4\right)^3 - 2 = 8 - 2$
10. $y = \frac{1}{x-2} + 5$; vertical asymptote $x = 2$, horizontal asymptote $y = 5$

Station 3: Piecewise Functions, Operations, and Composition

1. $f(-3) = 1$, $f(-1) = 1$, $f(2) = 4$, $f(3) = 3$
   check which inequality each input satisfies
2. $f(-1) = -1$, $f(1) = 2$, $f(2) = 5$; not continuous: it jumps at $x = 1$ (open circle $(1, 3)$, closed $(1, 2)$)
3. $g(x) = -x + 4$ for $x < 3$; $g(x) = x - 2$ for $x \geq 3$
   $|x - 3| = 3 - x$ *when* $x < 3$
4. $(f + g)(x) = x^2 + 3x + 1$; $(f \cdot g)(2) = 24$
   $f(2) = 3$ *and* $g(2) = 8$
5. $(f \circ g)(x) = x^2 - 6x + 8$; $(g \circ f)(x) = x^2 + 2x - 4$
   $f(g(x)) = (x-4)^2 + 2(x-4)$
6. $\sqrt{x^2 - 9}$; domain $(-\infty, -3] \cup [3, \infty)$
   need $x^2 - 9 \geq 0$
7. 4, 2, 4
   $f(g(3)) = f(1)$*;* $g(f(1)) = g(4)$*;* $f(f(2)) = f(1)$
8. $A(r(t)) = 9\pi t^2$; area $= 144\pi$ cm^2
   $r(4) = 12$ *and* $\pi(12)^2$
9. $K(C(F)) = \frac{5}{9}(F - 32) + 273.15$; $K(C(212)) = 373.15$ kelvins
   water boils at $212°F$
10. $f(x) = x^4$ and $g(x) = 5x - 2$
    inside function first (answers may vary)

Station 4: Inverse Functions

1. $f^{-1}(x) = \frac{x+7}{3}$
   swap x *and* y*, then solve for* y
2. $f^{-1}(x) = \frac{3x+1}{x-2}$
   $xy - 3x = 2y + 1 \Rightarrow y(x-2) = 3x + 1$
3. $f^{-1}(x) = \sqrt{x-1} + 2$; domain $x \geq 1$
   range of f *is* $y \geq 1$
4. $f(g(x)) = (\sqrt[3]{x-1})^3 + 1 = x$ and $g(f(x)) = \sqrt[3]{x^3} = x$
5. 1, 2, 4
   swap inputs and outputs
6. No: any line $y = k$ with $-2 < k < 2$ meets it 3 times; for example $y = 0$ at $x = -\sqrt{3},\ 0,\ \sqrt{3}$
7. $(0, -2)$, $(1, -1)$, $(2, 2)$; $f^{-1}(x) = x^2 - 2$, domain $x \geq 0$
   reflect in $y = x$*: swap coordinates*
8. $c^{-1}(x) = \frac{x - 2.50}{1.60}$; $c^{-1}(18.50) = 10$ miles
   the inverse takes a cost and returns miles
9. f^{-1} means the inverse, not the reciprocal. $f^{-1}(x) = \frac{x-6}{2}$, so $f^{-1}(4) = -1$, but $\frac{1}{f(4)} = \frac{1}{14}$
10. f: domain $[1, \infty)$, range $[3, \infty)$; f^{-1}: domain $[3, \infty)$, range $[1, \infty)$
    domain and range swap